Solid State Physics Tutorial

Energy Levels, Band Structure, and Effective Mass

Contents

1 Hydrogen Atom Energy Levels

1.1 Conceptual Questions

1. **Question 1.1:** What is the physical significance of the Bohr radius a_0 in the hydrogen atom, and how does it relate to the uncertainty principle?

   Answer 1.1: The Bohr radius ($a_0 = 0.529$) represents the characteristic size of the hydrogen atom and the most probable distance for finding an electron in the ground state. It emerges from balancing the Coulomb attraction (which tries to pull the electron into the nucleus) with the uncertainty principle (which prevents the electron from being localized too tightly). The uncertainty principle states $\Delta x \cdot \Delta p \geq \hbar/2$. If we confine the electron to distances $\sim a_0$, the momentum uncertainty becomes $\Delta p \geq \hbar/a_0$, giving kinetic energy $\sim \hbar^2/(2ma_0^2)$. Minimizing the total energy (kinetic + potential) yields $a_0 = 4\pi\epsilon_0\hbar^2/(m_e e^2)$.

2. **Question 1.2:** Explain why the energy levels of the hydrogen atom are negative, and what does $E = 0$ represent?

   Answer 1.2: Energy levels are negative because the electron is bound to the nucleus by the Coulomb attraction. We set the zero of potential energy at infinity (where the electron and proton are infinitely separated and at rest). When the electron is at any finite distance, it has lower total energy than at infinity because binding energy has been released. The energy $E = 0$ represents the ionization thresholdthe point at which the electron is just barely bound, with zero kinetic energy at infinite separation. Any state with $E < 0$ is bound; any state with $E > 0$ represents a free electron in the continuum.

3. **Question 1.3:** How does the energy difference between adjacent levels change as we go from $n = 1$ to higher quantum numbers, and what does this tell us about ionization?

   Answer 1.3: The energy difference $\Delta E_{n\to n+1} = E_{n+1} - E_n$ decreases rapidly as n increases. For example: $\Delta E_{1\to 2} = 10.2$ eV, $\Delta E_{2\to 3} = 1.89$ eV, $\Delta E_{3\to 4} = 0.85$ eV. As $n \to \infty$, $\Delta E \to 0$. This tells us that: (1) it takes much more energy to ionize from the ground state than from excited states, (2) highly excited atoms are easily ionized by small perturbations, and (3) there is a dense spectrum of levels just below the ionization threshold. This explains why stars and plasmas show ionization from excited states more easily than from ground states, and why atomic spectroscopy reveals the detailed structure of high-n levels in absorption spectra.

1.2 Numerical Exercises

Exercise 1.1: Energy Transition in Hydrogen

Problem: An electron in a hydrogen atom is in the $n = 3$ state. It transitions to the $n = 2$ state by emitting a photon. Calculate:

(a) The wavelength of the emitted photon

(b) The energy of the photon in eV

(c) In which region of the electromagnetic spectrum does this photon lie?

Given Data:

- Rydberg constant: $R_H = 1.097 \times 10^7\,\text{m}^{-1}$
- Planck constant: $h = 6.626 \times 10^{-34}\,\text{Js}$
- Speed of light: $c = 3.0 \times 10^8\,\text{m/s}$
- $1\,\text{eV} = 1.602 \times 10^{-19}\,\text{J}$

Solution:
Step 1: Recall the Rydberg formula for hydrogen
The wavelength of radiation emitted during a transition between levels n_i and n_f is given by:

$$\frac{1}{\lambda} = R_H \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right) \tag{1}$$

Explanation: This formula comes from quantizing angular momentum ($L = n\hbar$) and balancing the Coulomb force with centripetal force. The Rydberg constant encompasses the fundamental atomic parameters.
Step 2: Apply the formula with $n_i = 3$ and $n_f = 2$

$$\frac{1}{\lambda} = 1.097 \times 10^7 \left(\frac{1}{2^2} - \frac{1}{3^2} \right) \tag{2}$$

$$\frac{1}{\lambda} = 1.097 \times 10^7 \left(\frac{1}{4} - \frac{1}{9} \right) = 1.097 \times 10^7 \,(0.25 - 0.1111) \tag{3}$$

$$\frac{1}{\lambda} = 1.097 \times 10^7 \times 0.1389 = 1.523 \times 10^6\,\text{m}^{-1} \tag{4}$$

Meaning: The inverse wavelength tells us how many wavelengths fit in one metera larger value means shorter wavelength and higher energy photon.
Step 3: Calculate the wavelength

$$\lambda = \frac{1}{1.523 \times 10^6} = 6.566 \times 10^{-7}\,\text{m} = \boxed{656.6\,\text{nm}} \tag{5}$$

Meaning: This is the famous H_α line, a red emission line in the Balmer series. This transition occurs in hydrogen discharge lamps and neon signs.
Step 4: Calculate the photon energy

$$E_{\text{photon}} = h\nu = \frac{hc}{\lambda} \tag{6}$$

$$E_{\text{photon}} = \frac{6.626 \times 10^{-34} \times 3.0 \times 10^{8}}{6.566 \times 10^{-7}} = 3.024 \times 10^{-19}\,\text{J} \tag{7}$$

Step 5: Convert to eV

$$E_{\text{photon}} = \frac{3.024 \times 10^{-19}}{1.602 \times 10^{-19}} = \boxed{1.89\,\text{eV}} \tag{8}$$

Step 6: Identify the electromagnetic region

With wavelength $\lambda = 656.6$ nm, this photon lies in the **visible region** of the electromagnetic spectrum, specifically in the $\boxed{\text{red region}}$ (wavelengths 620-750 nm).

Physical Insight: The energy quantization in the hydrogen atom creates discrete spectral lines. The transition from $n = 3 \to n = 2$ represents the release of energy as an electron drops to a lower orbital. This is why hydrogen discharge tubes produce characteristic colors at specific wavelengthsa key experimental verification of quantum mechanics.

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Exercise 1.2: Ionization from Excited State

Problem: An electron in a hydrogen atom is in the $n = 2$ excited state. Calculate:

(a) The binding energy of the electron in this state

(b) The minimum energy required to ionize the atom from this state

(c) How many times more energy is needed to ionize from $n = 2$ compared to ionizing from $n = 1$?

Given: The ground state energy of hydrogen is $E_1 = -13.6$ eV

Solution:

Step 1: Recall the energy level formula for hydrogen

The energy of the n-th level in hydrogen is:

$$E_n = -\frac{13.6\,\text{eV}}{n^2} \tag{9}$$

Explanation: This comes from solving the Schrdinger equation for the Coulomb potential. The energy depends only on the principal quantum number n, not on l or m (this degeneracy is specific to hydrogen and hydrogen-like ions).

Step 2: Calculate the energy of the $n = 2$ state

$$E_2 = -\frac{13.6}{2^2} = -\frac{13.6}{4} = \boxed{-3.4\,\text{eV}} \tag{10}$$

Meaning: The negative sign indicates that the electron is bound. The magnitude represents the binding energy.

Step 3: Binding energy from $n = 2$

$$\text{Binding Energy} = |E_2| = \boxed{3.4\,\text{eV}} \tag{11}$$

Explanation: This is how much energy we must supply to remove the electron from the $n = 2$ state (bringing it to the ionization continuum at $E = 0$).
Step 4: Ionization energy from $n = 2$

$$\text{Ionization Energy from } n = 2 = E_\infty - E_2 = 0 - (-3.4) = \boxed{3.4\,\text{eV}} \tag{12}$$

Step 5: Compare ionization energies
From $n = 1$:

$$\text{Ionization Energy from } n = 1 = 0 - E_1 = 0 - (-13.6) = 13.6\,\text{eV} \tag{13}$$

Ratio:

$$\text{Ratio} = \frac{13.6}{3.4} = \boxed{4} \tag{14}$$

Physical Insight: It takes 4 times more energy to ionize from the ground state than from $n = 2$. This is why excited atoms are easier to ionizeelectrons are already partially "lifted" from the nucleus and experience less binding. This principle is crucial in plasma physics and stellar atmospheres.

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Exercise 1.3: Wave Function Probability

Problem: For a hydrogen atom in the ground state ($n = 1$, $l = 0$), the probability of finding the electron within a distance r from the nucleus is related to the Bohr radius $a_0 = 0.529$.

(a) What is the most probable radius (position) where the electron is found?

(b) Calculate the probability of finding the electron within one Bohr radius

(c) What is the physical significance of the Bohr radius in this context?

Given:

- Bohr radius: $a_0 = 0.529 \times 10^{-10}$ m
- For the 1s orbital: $R_{10}(r) = 2a_0^{-3/2}e^{-r/a_0}$
- Radial probability density: $P(r) = |R(r)|^2 r^2$

Solution:
Step 1: Identify the radial wave function
For the ground state of hydrogen:

$$R_{10}(r) = 2a_0^{-3/2}e^{-r/a_0} \tag{15}$$

Explanation: This is the radial part of the wave function. The full wave function is $\psi(r, \theta, \phi) = R(r)Y(\theta, \phi)$. Here Y = constant (spherically symmetric).
Step 2: Find the most probable radius
The radial probability density is:

$$P(r) = |R_{10}(r)|^2 r^2 = \left(2a_0^{-3/2}e^{-r/a_0}\right)^2 r^2 = 4a_0^{-3}r^2 e^{-2r/a_0} \tag{16}$$

To find the maximum, set $\frac{dP}{dr} = 0$:

$$\frac{dP}{dr} = 4a_0^{-3}\left(2re^{-2r/a_0} + r^2 \cdot \left(-\frac{2}{a_0}\right)e^{-2r/a_0}\right) = 0 \tag{17}$$

$$2r - \frac{2r^2}{a_0} = 0 \quad \Rightarrow \quad r = \boxed{a_0} \tag{18}$$

Meaning: The most probable distance to find the electron is exactly the Bohr radius! This is not by accidentit emerges naturally from the wave function.

Step 3: Probability within one Bohr radius

$$P(r \leq a_0) = \int_0^{a_0} 4a_0^{-3}r^2e^{-2r/a_0}dr \tag{19}$$

Let $u = \frac{2r}{a_0}$, so $dr = \frac{a_0}{2}du$:

$$P = 4a_0^{-3}\int_0^2 \frac{a_0^2u^2}{4}e^{-u}\frac{a_0}{2}du = \frac{1}{2}\int_0^2 u^2e^{-u}du \tag{20}$$

Using integration by parts twice:

$$\int_0^2 u^2e^{-u}du = \left[-u^2e^{-u} - 2ue^{-u} - 2e^{-u}\right]_0^2 = -(4e^{-2} + 4e^{-2} + 2e^{-2}) + 2 \tag{21}$$

$$= 2 - 10e^{-2} = 2 - 10(0.1353) = 2 - 1.353 = 0.647 \tag{22}$$

$$P(r \leq a_0) = \frac{1}{2} \times 0.647 = \boxed{0.323 \text{ or } 32.3\%} \tag{23}$$

Physical Insight: Only about 32

Step 4: Significance of the Bohr radius

$$a_0 = \frac{4\pi\epsilon_0\hbar^2}{m_ee^2} = \boxed{0.529} \tag{24}$$

This radius represents:

- The characteristic scale of the hydrogen atom
- The most probable distance from the nucleus (radial expectation value maximum)
- The length scale where quantum effects become dominant
- The balancing point between kinetic and potential energy

2 Kronig-Penney Model for Band Structure

2.1 Conceptual Questions

1. **Question 2.1:** What is the fundamental difference between the band structure produced by the Kronig-Penney model and a free electron gas, and why does the periodic potential create bands?

   Answer 2.1: In a free electron gas, the density of states is continuous and follows $g(E) \propto \sqrt{E}$. The Kronig-Penney model introduces a periodic potential, which creates band structurealternating regions of allowed energies (bands) separated by forbidden energies (band gaps). This occurs because: (1) At certain k-values (Brillouin zone boundaries), the electron wavelength matches the lattice periodicity, causing Bragg reflection and creating standing waves; (2) These standing waves have higher (or lower) potential energy depending on where they concentrate relative to the potential wells; (3) This splits the energy levels into two groups, creating a gap. The periodic potential effectively modulates the electron motion, preventing certain energy ranges from being accessible.

2. **Question 2.2:** Explain the concept of Brillouin zones and how they relate to reciprocal space. Why do we only need to consider the first Brillouin zone?

   Answer 2.2: The Brillouin zones are cells in reciprocal space (k-space) defined by perpendicular bisectors of reciprocal lattice vectors. The first Brillouin zone (1BZ) is the Wigner-Seitz cell of the reciprocal lattice. In 1D with lattice constant d, the 1BZ extends from $k = -\pi/d$ to $k = +\pi/d$. We only need to consider the 1BZ because of the periodicity of the band structure: states with k outside the 1BZ are equivalent to states inside by the addition of reciprocal lattice vectors. This is called the "band structure periodicity." Physically, it means the electron properties repeat with period $2\pi/d$ in k-space, so all unique information is contained within the 1BZ.

3. **Question 2.3:** What does the band gap represent physically, and under what conditions in the Kronig-Penney model does it become larger?

   Answer 2.3: A band gap is a range of forbidden energieselectrons cannot have energies within this range in an infinite, perfect crystal. Physically, it represents: (1) Regions where quantum interference from the periodic potential prevents wave propagation, (2) The energy cost to create a localized defect state, (3) The threshold energy for exciting electrons from valence to conduction bands. In the Kronig-Penney model, the band gap becomes larger when: (1) The potential strength V_0 increasesstronger potentials create stronger Bragg reflections, (2) The barrier width b increasesstronger barriers block more strongly, (3) The lattice constant d decreasestighter packing increases the scattering effect. Quantitatively: $E_g \propto V_0 \cdot (b/d)$ approximately. This explains why metals have small or zero band gaps (weak effective potential) while insulators have large band gaps (strong potential from core electrons).

2.2 Qualitative Analysis Exercise

Exercise 2.1: The Kronig-Penney Potential and Band Structure

Problem:

Consider a one-dimensional periodic potential used in the Kronig-Penney model:

$$V(x) = \begin{cases} 0 & \text{for } 0 < x < a \\ V_0 & \text{for } a < x < a+b \end{cases} \tag{25}$$

with period $d = a + b$.

(a) Sketch the band structure (E vs. k) for the case where the potential strength V_0 is small

(b) Sketch the band structure for the case where V_0 is large

(c) Explain what happens to the band gap at the Brillouin zone boundary

Solution:

Background: The Kronig-Penney Equation

In the Kronig-Penney model, we solve the time-independent Schrdinger equation in a periodic potential:

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} + V(x)\psi = E\psi \tag{26}$$

By Bloch's theorem, the wave function has the form:

$$\psi(x) = e^{ikx}u(x) \tag{27}$$

where $u(x)$ has the same periodicity as $V(x)$.

Step 1: Understand the physics in the free electron region

When $V_0 = 0$ (no potential), we get a parabolic band $E(k) = \frac{\hbar^2 k^2}{2m}$. All states are allowedthere are no gaps.

Step 2: Weak potential case (V_0 small)

When a weak periodic potential is introduced:

- The band structure remains nearly parabolic
- Small gaps (band gaps) appear only at the Brillouin zone boundaries where $k = \pm\frac{n\pi}{d}$
- The gaps grow like $\propto V_0$ (first order)
- Multiple bands form, but they are well-separated

Sketch for weak potential:

```
E
^
|     Band 2 (nearly parabolic)
|    /   \
|   /  gaps (small)
|  /       \
|_/_ _ _ _\ _ _ _ k
  -pi/d 0 pi/d

  Band 1
```

Step 3: Strong potential case (V_0 large)
When the potential is strong:

- Band gaps become very large
- Bands become flatter (less dispersive)
- The energy ranges might even separate into distinct, non-overlapping bands
- Near band edges, the $E(k)$ is nearly flat (constant energy)

Sketch for strong potential:

```
E
^
|     _____ Band 2 (flat)
|    |
|    | Large gap
|    |_____ Band 1 (flat)
|____|_______ k
  -pi/d  0  pi/d
```

Step 4: Understanding band gaps at the Brillouin zone boundary
At the Brillouin zone boundary ($k = \pi/d$), there is special significance:

$$k = \pm\frac{n\pi}{d} \tag{28}$$

These k-values correspond to standing wave patterns in the periodic lattice (Bragg scattering condition: $2d\sin\theta = n\lambda$).

When electron waves scatter constructively from the periodic potential, they can create regions of forbidden energy (band gaps).

Physical Insight: The Kronig-Penney model shows that: 1. A periodic potential naturally creates band structure from a continuous spectrum 2. The strength of the potential determines the size of the band gap 3. The periodicity determines where the gaps occur (at specific k-values)

This is why conductors have overlapping bands (small gaps or no gaps) while insulators have large gapsit's a consequence of the atomic potential strength and spacing!

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Exercise 2.2: Brillouin Zone and the Density of States

Problem:

In the Kronig-Penney model with lattice constant $d = 3.0$:

(a) Calculate the boundaries of the first Brillouin zone

(b) Explain why the density of states is different inside and outside a band

(c) Why is the density of states infinite at the band edges?

Solution:
Step 1: Define the Brillouin zone
The first Brillouin zone (1BZ) in one dimension is defined as the Wigner-Seitz cell in reciprocal space:

$$\text{1BZ boundaries: } k = \pm\frac{\pi}{d} \tag{29}$$

Step 2: Calculate the boundaries
Given $d = 3.0 \ = 3.0 \times 10^{-10}$ m:

$$k_{\text{boundary}} = \pm\frac{\pi}{d} = \pm\frac{\pi}{3.0 \times 10^{-10}} = \pm\boxed{1.047 \times 10^{10}\,\text{m}^{-1}} \tag{30}$$

Physical Meaning: All electrons in the periodic potential can be described using k-values within this range. States outside this range are equivalent to states inside (they differ only by a reciprocal lattice vector).
Step 3: Density of states in a band
The density of states $g(E)$ in k-space is approximately uniform (in 1D), but in real space it must account for the band curvature:

$$g(E) = \frac{L}{\pi}\left|\frac{dk}{dE}\right| \tag{31}$$

where L is the system length.
Inside a band (away from edges), $E(k)$ varies smoothly, so $|dk/dE|$ is roughly constant $g(E)$ is roughly constant.
Step 4: Density of states at band edges
At a band edge, the band curves with minimum curvature:

$$E(k) \approx E_{\text{edge}} + \alpha(k - k_{\text{edge}})^2 \tag{32}$$

Therefore:

$$\frac{dE}{dk} \approx 0 \quad \Rightarrow \quad \left|\frac{dk}{dE}\right| \to \infty \tag{33}$$

Result:

$$g(E) \to \infty \text{ at band edges} \tag{34}$$

Physical Significance:

- Van Hove singularity: DOS diverges at band edges
- This explains why electrons accumulate near band edges
- Important for semiconductors: the band edge region determines most properties
- Explains why impurities/defects near band edges have large effects

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Exercise 2.3: Band Gap Calculation in the Weak Potential Limit

Problem:

In the weak potential limit of the Kronig-Penney model, the band gap at the Brillouin zone boundary $k = \pi/d$ can be approximated as:

$$E_g = \frac{2V_0 a}{d} \tag{35}$$

Given:

- Potential well width: $a = 2.0$
- Potential barrier height: $V_0 = 2.0$ eV
- Period: $d = 3.0$

(a) Calculate the band gap

(b) If we double the potential strength ($V_0 \to 2V_0$), how does the band gap change?

(c) If we double the lattice constant ($d \to 2d$), how does the band gap change?

Solution:

Step 1: Calculate the band gap

Using the weak potential approximation:

$$E_g = \frac{2V_0 a}{d} = \frac{2 \times 2.0\,\text{eV} \times 2.0}{3.0} \tag{36}$$

$$E_g = \frac{8.0}{3.0}\,\text{eV} = \boxed{2.67\,\text{eV}} \tag{37}$$

Meaning: This is the energy range forbidden to electrons at the Brillouin zone boundary. Electrons cannot have energies in this range near $k = \pi/d$.

Step 2: Effect of doubling potential strength

$$E'_g = \frac{2(2V_0)a}{d} = \frac{4V_0 a}{d} = 2E_g = \boxed{5.34\,\text{eV}} \tag{38}$$

Scaling: $E_g \propto V_0$ (linear dependence)

Physical Insight: A stronger periodic potential creates larger band gaps. This is why: - Tight-binding materials have larger band gaps - Loosely bound electrons (weak potential) have small gaps - Hard-core repulsive potentials create wide band gaps

Step 3: Effect of doubling lattice constant

$$E''_g = \frac{2V_0 a}{2d} = \frac{V_0 a}{d} = \frac{E_g}{2} = \boxed{1.33\,\text{eV}} \tag{39}$$

Scaling: $E_g \propto 1/d$ (inverse dependence)

Physical Insight: Larger lattice constant means: - Atoms farther apart - Weaker periodic potential effect - Smaller band gaps - More free-electron-like behavior

This explains why: - Bulk semiconductors have smaller gaps than quantum dots (confined geometries have smaller d) - Light elements in dense solids have large band gaps - Loosely packed materials have small band gaps

Summary Table:

Change	**Band Gap**
Increase V_0	E_g increases
Increase d	E_g decreases
Increase a	E_g increases

3 Effective Mass Calculation for Electrons and Holes

3.1 Conceptual Questions

1. **Question 3.1:** What is meant by "effective mass" and why is it different from the free electron mass? Under what conditions can it become negative or infinite?

   Answer 3.1: Effective mass m^* is the mass parameter that appears in the equation of motion for an electron in a crystal: $m^* \frac{d\vec{v}}{dt} = \vec{F}$. It accounts for the combined effects of the true electron mass and the periodic crystal potential. It differs from the free electron mass because: (1) The crystal potential accelerates or decelerates the electron (changing its effective inertia), (2) The band structure curvature determines the response to external forces. Mathematically, $m^* = \hbar^2 (d^2E/dk^2)^{-1}$. It can be: **Negative** at the top of a valence band where $d^2E/dk^2 < 0$ (this represents holes), **Very large** near a band edge where $d^2E/dk^2 \to 0$ (the band becomes flat, making acceleration difficult), **Infinite** exactly at the band maximum or minimum where the curvature vanishes (the electron cannot accelerate, like a particle at the top of a potential hill).

2. **Question 3.2:** Explain the physical difference between electrons near the bottom of a band and holes near the top of a band, using the concept of effective mass.

   Answer 3.2: Near the band bottom (conduction band minimum): electrons have positive effective mass $m_e^* > 0$, positive curvature of the band ($d^2E/dk^2 > 0$), and respond to force with acceleration in the expected direction. They're similar to free particles, just with modified inertia. Near the band top (valence band maximum): missing electrons (holes) effectively have negative curvature ($d^2E/dk^2 < 0$), leading to positive effective mass $m_h^* > 0$ when we account for their positive charge. Crucially, a hole accelerates in the direction of the applied electric field (like a positive charge), while an electron accelerates opposite to the field. This happens because at the band top, increasing k actually decreases energy. The band curvature essentially reverses the dynamics: the periodic potential now "helps" the particle move, while electrons at the band bottom work "against" it. This counterintuitive behavior is why holes are not just "missing electrons" but genuinely different quasiparticles with reversed dynamics.

3. **Question 3.3:** Why is the concept of effective mass so powerful in semiconductor physics, and what limitations does it have?

   Answer 3.3: Effective mass is powerful because it allows us to use simple Newtonian mechanics to describe electrons in crystals: $F = m^* a$. Instead of solving the full Schrdinger equation with a complicated periodic potential, we can use parabolic band approximation and treat electrons like classical particles with modified mass. This simplification enables: (1) Quick estimates of mobility, conductivity, and device performance, (2) Understanding of semiconductor band structure through simple numbers, (3) Design of devices based on classical reasoning. However, limitations include: (1) It only works near band extrema where $E(k)$ is approximately parabolicfails for large k or in the middle of bands, (2) It assumes single-particle picture and ignores electron-electron interactions and scattering, (3) It doesn't account for nonparabolicity (deviations from quadratic dispersion), (4) It breaks down under very strong electric fields (breakdown of band picture), (5) For highly anisotropic materials, using scalar m^* can be misleadingneed the full tensor. Despite these limitations, effective mass remains the foundation of semiconductor device physics because it captures the essential physics in a transparent, calculable way.

3.2 Numerical Exercises

Exercise 3.1: Effective Mass from Band Curvature

Problem:

For a parabolic band approximation, the energy dispersion near the band minimum is:

$$E(k) = E_c + \frac{\hbar^2 k^2}{2m^*} \tag{40}$$

where m^* is the effective mass, E_c is the conduction band minimum, and k is the wave vector measured from the band center.

For a particular semiconductor, experimental band structure measurements give:

- At $k = 0.1 \times 10^{10}$ m^{-1}: $E = E_c + 0.05$ eV
- At $k = 0.2 \times 10^{10}$ m^{-1}: $E = E_c + 0.20$ eV

(a) Calculate the effective mass

(b) Express it as a fraction of the free electron mass

(c) Discuss what this value tells us about electron behavior in this semiconductor

Solution:

Step 1: Set up the effective mass equation

The band dispersion near the minimum is:

$$E(k) - E_c = \frac{\hbar^2 k^2}{2m^*} \tag{41}$$

Explanation: This is analogous to the free electron kinetic energy, but with m^* replacing the free electron mass. This parabolic approximation is valid near the band extremum where the band is smooth and approximately quadratic.

Step 2: Use the first data point

$$0.05\,\text{eV} = \frac{\hbar^2 (0.1 \times 10^{10})^2}{2m^*} \tag{42}$$

Step 3: Use the second data point

$$0.20\,\text{eV} = \frac{\hbar^2 (0.2 \times 10^{10})^2}{2m^*} \tag{43}$$

Step 4: Take the ratio to eliminate m^*

$$\frac{0.20}{0.05} = \frac{(0.2)^2}{(0.1)^2} = \frac{0.04}{0.01} = 4 \tag{44}$$

This is consistent! Now solve for m^*:

$$0.05\,\text{eV} = \frac{\hbar^2 (0.1 \times 10^{10})^2}{2m^*} \tag{45}$$

Convert $\hbar$ and energies to SI units:

- $\hbar = 1.055 \times 10^{-34}$ Js

- $0.05 \text{ eV} = 0.05 \times 1.602 \times 10^{-19} \text{ J} = 8.01 \times 10^{-21} \text{ J}$
- $k = 0.1 \times 10^{10} \text{ m}^{-1} = 1.0 \times 10^{9} \text{ m}^{-1}$
- Free electron mass: $m_e = 9.109 \times 10^{-31}$ kg

Step 5: Solve for m^*

$$m^* = \frac{\hbar^2 k^2}{2(E - E_c)} \tag{46}$$

$$m^* = \frac{(1.055 \times 10^{-34})^2 \times (1.0 \times 10^{9})^2}{2 \times 8.01 \times 10^{-21}} \tag{47}$$

$$m^* = \frac{1.113 \times 10^{-68} \times 1.0 \times 10^{18}}{1.602 \times 10^{-20}} \tag{48}$$

$$m^* = \frac{1.113 \times 10^{-50}}{1.602 \times 10^{-20}} = 6.95 \times 10^{-31} \,\text{kg} \tag{49}$$

Step 6: Express as fraction of m_e

$$\frac{m^*}{m_e} = \frac{6.95 \times 10^{-31}}{9.109 \times 10^{-31}} = \boxed{0.76\, m_e} \tag{50}$$

Meaning: The effective mass is about 76
Step 7: Physical significance

1. **Lighter than free electron:** The effective mass $m^* < m_e$ means electrons in this band respond faster to electric fields and accelerate more easily
2. **Band structure effect:** The reduced mass is not a fundamental property of the electron, but rather emerges from the periodic potential. The periodic atomic potential actually *helps* the electron moveit's easier to move through a regular crystal than through vacuum
3. **Quantum origin:** The band curvature reflects how the periodic potential modulates the electron's kinetic and potential energy. A lighter effective mass indicates a "flatter," more dispersive band where electrons gain speed quickly with increasing k
4. **Material property:** This measured value is characteristic of the semiconductor material and depends on the strength and periodicity of the atomic potential. Different materials have different effective masses because their band structures are different

Conclusion: In this semiconductor, electrons move with an effective inertia about 76
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Exercise 3.2: Holes and Effective Mass at Band Maximum

Problem:

Near the top of a valence band in a semiconductor, the energy dispersion is:

$$E(k) = E_v - \frac{\hbar^2 k^2}{2m_h^*} \tag{51}$$

where E_v is the valence band maximum and m_h^* is the hole effective mass. Experimental measurements show:

- At $k = 0$: $E = E_v$
- At $k = 0.1 \times 10^{10}$ m^{-1}: $E = E_v - 0.08$ eV

(a) Calculate the hole effective mass

(b) Compare it with the electron effective mass from Exercise 3.1

(c) Explain the physical difference between the band curvature at the top versus the bottom of a band

Solution:

Step 1: Understand the valence band curvature

At the top of a band, the energy decreases away from the center (k=0). This inverted curvature is described by:

$$E(k) = E_v - \frac{\hbar^2 k^2}{2m_h^*} \tag{52}$$

The second derivative is:

$$\frac{d^2E}{dk^2} = -\frac{\hbar^2}{m_h^*} < 0 \tag{53}$$

Explanation: The negative curvature at the band top is what makes holes behave differently from electrons.

Step 2: Calculate the hole effective mass

From the valence band equation:

$$E_v - 0.08\,\text{eV} = E_v - \frac{\hbar^2 (0.1 \times 10^{10})^2}{2m_h^*} \tag{54}$$

$$0.08\,\text{eV} = \frac{\hbar^2 (1.0 \times 10^9)^2}{2m_h^*} \tag{55}$$

$$m_h^* = \frac{\hbar^2 k^2}{2 \times 0.08\,\text{eV}} = \frac{(1.055 \times 10^{-34})^2 (1.0 \times 10^9)^2}{2 \times 0.08 \times 1.602 \times 10^{-19}} \tag{56}$$

$$m_h^* = \frac{1.113 \times 10^{-50}}{2.562 \times 10^{-20}} = 4.34 \times 10^{-31}\,\text{kg} \tag{57}$$

$$\frac{m_h^*}{m_e} = \frac{4.34 \times 10^{-31}}{9.109 \times 10^{-31}} = \boxed{0.48\,m_e} \tag{58}$$

Physical Insight: Holes often have smaller effective mass than electrons because they are near the band top where the curvature is sharp.

Step 3: Why holes behave as positive charges

Consider an electron in a nearly-full valence band:

Physical Picture: When we remove an electron from the valence band (creating a hole), we're removing a particle with negative charge. From the perspective of the remaining electrons, this absence acts like a positive charge. But more importantly, the *dynamics* are reversed:

When an electric field $\vec{E}$ is applied:

Electron in field:

$$\vec{F}_e = -e\vec{E} \quad \text{(force opposite to field)} \tag{59}$$

Hole (missing electron) in field:

Because the band has inverted curvature at its maximum, when the electron is removed, the remaining system behaves as if a positive particle were accelerating in the field direction:

$$\vec{F}_h = +e\vec{E} \quad \text{(force in direction of field)} \tag{60}$$

Explanation: This happens because the hole concept represents the collective behavior of many electrons in a nearly-full band. When one electron moves, all the others rearrange, and the net effect is equivalent to a positive charge moving in the opposite direction. The band curvature at the top (concave downward) ensures this effective positive charge behavior.

Step 4: Comparison of band structures

Property	**Conduction Band**	**Valence Band**
Energy relation	$E = E_c + \frac{\hbar^2 k^2}{2m_e^*}$	$E = E_v - \frac{\hbar^2 k^2}{2m_h^*}$
Band curvature	$\frac{d^2E}{dk^2} > 0$ (concave up)	$\frac{d^2E}{dk^2} < 0$ (concave down)
Carrier	Electron	Hole (missing e^-)
Charge	$-e$	$+e$
Force direction	Opposite to $\vec{E}$	Same as $\vec{E}$

Physical Insight: The inverted curvature at the band top is crucial. It means that as we increase k (move away from the band center), the energy *decreases*. For electrons, this unusual dispersion at the band top is why they're better described as holespositive quasiparticles with opposite dynamics. This is not just semantic; it profoundly affects how holes and electrons respond to external forces and how they recombine.

Isotropic vs Anisotropic Materials

The effective mass, m^*, describes how an electron or hole responds to an external force inside a crystal. For isotropic materials, m^* is a scalar, but for anisotropic materials, it is a tensor, reflecting the directional dependence of the band curvature. This exercise explores these concepts.

A novel semiconductor material has been synthesized, and its conduction band edge near the Γ-point is described by the following energy–momentum (E–$\mathbf{k}$) dispersion relation:

$$E(\mathbf{k}) = E_c + \frac{\hbar^2}{2}\left(\frac{k_x^2}{m_x^*} + \frac{k_y^2}{m_y^*} + \frac{k_z^2}{m_z^*}\right)$$

where E_c is the energy at the conduction band minimum, $\hbar$ is the reduced Planck constant, and m_x^*, m_y^*, and m_z^* are the effective masses along the crystallographic axes. The material's properties are determined to be $m_x^* = 0.2m_0$, $m_y^* = 0.5m_0$, and $m_z^* = 0.3m_0$, where m_0 is the free electron mass.

Questions

1. **Effective Mass Tensor:** Write down the 3×3 effective mass tensor, $\mathbf{m}^*$, for this material.

2. **Acceleration in Response to an Electric Field:**

   (a) An electric field, $\mathbf{E} = E_0\hat{\mathbf{x}}$, is applied along the x-direction. Calculate the electron's acceleration vector, $\mathbf{a}$. Is the acceleration parallel to the applied force?

   (b) Now, an electric field, $\mathbf{E} = E_0(\hat{\mathbf{x}} + \hat{\mathbf{y}})$, is applied. Calculate the electron's acceleration vector, $\mathbf{a}$. Is the acceleration parallel to the applied force?

3. **Comparison with an Isotropic Material:**
   A second, isotropic semiconductor has a single, spherical conduction band valley at Γ. This material has an effective mass of $m^* = 0.2m_0$. An electric field, $\mathbf{E} = E_0(\hat{\mathbf{x}} + \hat{\mathbf{y}})$, is applied.

   (a) Calculate the electron's acceleration vector, $\mathbf{a}$.

   (b) Compare this result with your finding in Question 2b. Explain the difference in the acceleration vectors based on the material's isotropy versus anisotropy.

Key Formulas

$$\left(\frac{1}{\mathbf{m}^*}\right)_{ij} = \frac{1}{\hbar^2}\frac{\partial^2 E(\mathbf{k})}{\partial k_i \partial k_j}$$

$$\mathbf{a} = \left(\frac{1}{\mathbf{m}^*}\right) \cdot \mathbf{F}, \qquad \mathbf{F} = -e\mathbf{E}$$

Guided Solution to Effective Mass Exercise

1. Effective Mass Tensor

Concept: The effective mass tensor, $\mathbf{m}^*$, relates the applied force to the resulting acceleration. Its inverse is defined by the second derivative of energy with respect to momentum.

$$\left(\frac{1}{\mathbf{m}^*}\right)_{ij} = \frac{1}{\hbar^2}\frac{\partial^2 E}{\partial k_i \partial k_j}$$

For the given $E(\mathbf{k})$,

$$\frac{\partial^2 E}{\partial k_x^2} = \frac{\hbar^2}{m_x^*}, \quad \frac{\partial^2 E}{\partial k_y^2} = \frac{\hbar^2}{m_y^*}, \quad \frac{\partial^2 E}{\partial k_z^2} = \frac{\hbar^2}{m_z^*}$$

All mixed partial derivatives vanish. Therefore,

$$\left(\frac{1}{\mathbf{m}^*}\right) = \begin{pmatrix} 1/m_x^* & 0 & 0 \\ 0 & 1/m_y^* & 0 \\ 0 & 0 & 1/m_z^* \end{pmatrix}, \qquad \mathbf{m}^* = \begin{pmatrix} m_x^* & 0 & 0 \\ 0 & m_y^* & 0 \\ 0 & 0 & m_z^* \end{pmatrix} = m_0 \begin{pmatrix} 0.2 & 0 & 0 \\ 0 & 0.5 & 0 \\ 0 & 0 & 0.3 \end{pmatrix}.$$

2. Acceleration in Response to an Electric Field

Concept: The force on an electron is $\mathbf{F} = -e\mathbf{E}$. The acceleration is given by $\mathbf{a} = (\mathbf{m}^*)^{-1} \cdot \mathbf{F}$.

(a) For $\mathbf{E} = E_0\hat{\mathbf{x}}$:

$$\mathbf{F} = -eE_0\hat{\mathbf{x}} = (-eE_0, 0, 0)$$

$$\mathbf{a} = \begin{pmatrix} 1/m_x^* & 0 & 0 \\ 0 & 1/m_y^* & 0 \\ 0 & 0 & 1/m_z^* \end{pmatrix} \begin{pmatrix} -eE_0 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} -eE_0/m_x^* \\ 0 \\ 0 \end{pmatrix} = -\frac{eE_0}{0.2m_0}\hat{\mathbf{x}}.$$

The acceleration is parallel to the applied field.

(b) For $\mathbf{E} = E_0(\hat{\mathbf{x}} + \hat{\mathbf{y}})$:

$$\mathbf{F} = -eE_0(\hat{\mathbf{x}} + \hat{\mathbf{y}}) = (-eE_0, -eE_0, 0)$$

$$\mathbf{a} = \begin{pmatrix} 1/m_x^* & 0 & 0 \\ 0 & 1/m_y^* & 0 \\ 0 & 0 & 1/m_z^* \end{pmatrix} \begin{pmatrix} -eE_0 \\ -eE_0 \\ 0 \end{pmatrix} = \begin{pmatrix} -eE_0/m_x^* \\ -eE_0/m_y^* \\ 0 \end{pmatrix} = -eE_0\left(\frac{1}{0.2m_0}\hat{\mathbf{x}} + \frac{1}{0.5m_0}\hat{\mathbf{y}}\right).$$

Here, $\mathbf{a}$ is not parallel to $\mathbf{F}$ because $m_x^* \neq m_y^*$.

3. Comparison with an Isotropic Material

(a) For isotropic case ($m_x^* = m_y^* = m_z^* = 0.2m_0$):

$$\left(\frac{1}{\mathbf{m}^*}\right) = \begin{pmatrix} 1/0.2m_0 & 0 & 0 \\ 0 & 1/0.2m_0 & 0 \\ 0 & 0 & 1/0.2m_0 \end{pmatrix}$$

$$\mathbf{a} = \begin{pmatrix} 1/0.2m_0 & 0 & 0 \\ 0 & 1/0.2m_0 & 0 \\ 0 & 0 & 1/0.2m_0 \end{pmatrix} \begin{pmatrix} -eE_0 \\ -eE_0 \\ 0 \end{pmatrix} = -\frac{eE_0}{0.2m_0}(\hat{\mathbf{x}} + \hat{\mathbf{y}}).$$

Acceleration and force are perfectly parallel.

(b) Comparison:

- **Anisotropic material:** $\mathbf{a} = -eE_0 \left(\frac{1}{0.2m_0}\hat{\mathbf{x}} + \frac{1}{0.5m_0}\hat{\mathbf{y}} \right)$.
- **Isotropic material:** $\mathbf{a} = -\dfrac{eE_0}{0.2m_0}(\hat{\mathbf{x}} + \hat{\mathbf{y}})$.

In an anisotropic crystal, acceleration is not necessarily parallel to the applied force because the effective mass depends on direction. In isotropic materials, the effective mass is identical in all directions, so acceleration always follows the force.